

Iterated Prisoner's Dilemma

The lexicon, tiered

The branch index asks the question this project exists for.

*“how do humans manage multiple strategic relationships with their limited computing resources?
what can we learn from humans? how can we complement them?”*

website, Social World Modeling, the branch index

An agent playing one partner does not have this problem. An agent playing eight of them at once does, and it is a representation problem before it is a strategy problem.

“An agent that interacts with many partners simultaneously faces a representation problem that a dyadic learner does not: it must decide how much of its limited experience with any one partner to generalize toward others, and how much to keep partner-specific.”

ablation report, Introduction

Two animations on the page state the two halves. The first puts agents on a network and shows what one defector does to a neighbourhood. The second drops the network, takes one agent's point of view, and shows it sorting nine partners into three kinds. Between them sits the mechanism the page names and does not expand: the categories are relative, so the same partner is the exploiter in one room and the cooperator in another. That is contrast, and this file is where it gets its section.

Active entries carry all four fields, because those are the ones I have to define on the spot to a modeller, to a reviewer from psychology, and to a reviewer from multi-agent learning, who will each hear a different thing. Working entries carry a definition and the discrimination. Peripheral entries carry a definition and a pointer.

Six terms are used in two senses across my own files, and one of them is used in two senses on a single page. Each sits inside the entry it belongs to.

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1 Active

1.1 The game and the network

iterated prisoner's dilemma

“Each agent plays its neighbours repeatedly, and each carries a fixed strategy”

website, Iterated Prisoner's Dilemma, Dynamic Sociality

Definition: A two-choice mixed-motive game played repeatedly by the same pair, so that today's choice is evidence about tomorrow's partner. Here it is played on a graph, and each agent plays every neighbour rather than one opponent.

Distinguishes it from: The one-shot game, where defection is dominant and nothing is learned. And from the dyadic iterated game, where the partner is fixed and there is nothing to attribute: on a graph, an agent that was exploited has to decide who exploited it before it can respond.

Operational test: Does an agent face the same partner more than once, and does it face more than one partner? Both yes puts you in this setting; the second one no puts you in the dyadic literature, where most of the results were established.

In my project: Eight agents, twelve ties, fourteen rounds on the page's first animation. In the network simulations, groups of fifteen over one hundred rounds, and a triadic version in the incentives and information study.

fixed strategy

“Nobody sees the strategy list; they only see what happened last round.”

website, Iterated Prisoner's Dilemma, Dynamic Sociality

Definition: A partner policy that maps the last round to the next choice and never updates: tit-for-tat, always cooperate, always defect, grim. The population the learning agent has to read.

Distinguishes it from: A learning partner. A fixed strategy is stationary, so a category assigned to it stays correct; against a learner the category is a moving target and category recovery stops being well posed.

Operational test: Would the partner make the same choice in round ninety that it made in round ten, given the same last round? If yes it is fixed.

In my project: Five heuristics in the network code, spanning fully cooperative to fully defective: always cooperate, tit-for-tat, two probabilistic cooperators at seventy and fifty per cent, always defect. Grim is on the page's animation and not in that five.

tie

“A tit-for-tat agent next to a defector stops cooperating, and its other ties go cold too. The damage is not confined to the pair that caused it, because the retaliating agent cannot separate the neighbour who exploited it from the neighbours who didn't.”

website, Iterated Prisoner's Dilemma, Dynamic Sociality

Definition: An edge. One relationship, carrying its own history and, in the identity model, its

own memory and its own prosociality value.

Distinguishes it from: A round. A round is a slice of time across the whole graph; a tie is a slice of history between two agents, and the per-tie count of encounters is what the topology result turns on.

Operational test: Does the quantity have two agents in its index? Then it belongs to a tie and not to an agent.

In my project: Retaliation spills across ties because the retaliating agent does not attribute. That spillover is the argument for attribution, and it is why the second animation exists.

degree of interdependence

“The degree of interdependence k is the proportion of outcome variance attributable to the partner’s behavior”

Collective Intelligence manuscript, Degree of Interdependence

“When $k = 0$, outcomes are entirely self-determined; when $k = 1$, outcomes depend entirely on the partner’s actions”

same

Definition: $k = PC/(AC + PC)$, over the three components of interdependence theory: actor control, how much your own action determines your outcome; partner control, how much theirs does; joint control, how much the combination does.

Distinguishes it from: The payoff difference between cooperating and defecting, which is a magnitude. k is a ratio, and it is what decides whether network structure matters at all: at $k = 0.09$ all three topologies converge and the structure is irrelevant.

Operational test: Hold your own action fixed and vary the partner’s. How much of your outcome moved? That fraction is k .

In my project: Two levels instantiated through the matrix. High interdependence, $k = 1.00$, pays five for matching and zero otherwise. Low interdependence, $k = 0.09$, makes one action nearly dominant, and the topology effect disappears there.

topology

“This structure maximizes connectivity and partner diversity, but eliminates partner stability: an agent cannot reliably anticipate whom it will interact with next.”

Collective Intelligence manuscript, Network Structures

Definition: Three structures, ordered by connectivity. Disconnected dyads, one fixed partner each. Lattice, a ring with a few cross-links. Fully connected, every agent to every other, with the partner drawn at random each round.

Distinguishes it from: Group size, which is held constant at fifteen across the three. What varies is not how many agents there are but how the encounters are distributed among them.

Operational test: Can an agent predict who it will play next? Certainly in a dyad, approximately in a lattice, not at all in a fully connected network. That is the ordering that matters, not the edge count.

In my project: One hundred and five edges fully connected, seventeen to nineteen in the lattice, seven in the disconnected condition, where the fifteenth agent has no partner and does not play.

the connectivity paradox

“the connectivity paradox, in which more connections lead to worse collective outcomes, is specific to environments where successful coordination requires mutual adaptation with a consistent partner”

Collective Intelligence manuscript, Results

Definition: Under high interdependence, sparse networks outperform dense ones. Maximisation reaches .84 in disconnected dyads, .63 in the lattice and .51 fully connected, and the human experiment reproduces the ordering.

Distinguishes it from: The information-diffusion result it contradicts, where dense networks converge faster on a shared solution. The difference is the task: diffusion spreads an answer, coordination requires the other party to move at the same time.

Operational test: Does success require the partner to act with you, or only to have known something before you? Only the first produces the paradox.

In my project: Ruled out as an aggregation artifact by re-indexing per trial rather than per round, which holds observation count constant and leaves the dyadic advantage intact.

encounter frequency

“Identity provides the mechanism; encounter frequency provides the opportunity.”

Collective Intelligence manuscript, Full Information

“in a disconnected dyad network, each pair meets on every round, accumulating 100 encounters over 100 rounds. In the fully connected network with 15 agents, random matching yields approximately 7 encounters per pair over the same horizon.”

same

Definition: The number of times one particular pair has met. The index that makes the three topologies coincide: re-index coordination by per-pair encounter count and the curves collapse onto one.

Distinguishes it from: Experience, which is per agent and accumulates faster in dense networks. Per-pair encounters run the other way, and the second is what predicts coordination.

Operational test: Plot the outcome against rounds and against per-pair encounters. If the topologies separate on the first and coincide on the second, the effect was distributional, not cognitive.

In my project: This is the result that reinterprets the whole topology finding. Agents in all three structures learn at the same rate per encounter, so the sparse advantage is about how many encounters fit in the window and not about any structural benefit to learning.

win-stay lose-shift

“Win-stay therefore fails not because agents abandon the strategy, but because the network structure makes it incoherent: the partner required to complete the coordination is not guaranteed to be present.”

Definition: Repeat the last action after a rewarding outcome, switch after a poor one. The mechanism behind the dyadic advantage, and it needs no representation of the partner at all.

Distinguishes it from: Tit-for-tat, which conditions on the partner’s last action rather than on your own outcome. Tit-for-tat needs to see what the partner did; win-stay lose-shift needs only your own payoff, which is why it survives under minimal information.

Operational test: Classify each choice at t against the outcome at $t - 1$. Four cells: win-stay, win-change, lose-stay, lose-change. Adherence is the proportion in the two diagonal cells.

In my project: Sustained at .778 under individual memory in every topology and lower in every topology under pooled memory, with the widest gap fully connected. In the human experiment, win-stay use differs by condition, $F(2, 26) = 4.20$, $p = .026$.

1.2 The agent

instance

Definition: One stored experience. Under minimal information it is a pair, my action and my outcome. Under identity it is five slots: partner index, my action, the partner’s action, my outcome, the partner’s outcome.

Distinguishes it from: A state-action value. An instance is a specific past episode retrieved by similarity, not a running average; two agents with identical averages can behave differently because their instances differ in recency.

Operational test: Does the record name a particular occasion, or summarise many? Only the first is an instance.

In my project: The move from the two-slot to the five-slot instance is the whole difference between the minimal-information model and the identity model, and it is what makes a partner-specific memory possible.

blended value

$$“V_k(t) = \sum_{i=1}^n P_{ik}(x_{self} + \alpha(t) \cdot x_{other}) ”$$

Collective Intelligence manuscript, IBL Model with Identity

Definition: The value of an action, computed as a retrieval-probability weighted average over the instances consistent with it, with the partner’s outcome entering weighted by the prosociality value.

Distinguishes it from: An expected value under a learned distribution. Blending is over what was retrieved, so it inherits recency and frequency effects and the noise term, and two agents with the same history can choose differently.

Operational test: Would removing an old, rarely repeated experience change the number? Under blending it changes it a little; under a running average of the same data it changes it by a fixed share.

In my project: Decay 0.5, mismatch penalty 1, noise 0.25, the standard defaults, held across every condition so that differences are attributable to the mechanisms and not to the fit.

individual memory against pooled memory

“Individual memory outperformed pooled memory on every measured metric (cooperation rate, cumulative rewards, average rewards, and WSLS adherence) across all three network types.”

IBL-Network, Reports/2025_0722.md, summary

*“Pooled memory creates a **strategic averaging problem**. Instead of learning ‘cooperate with cooperators, defect against defectors,’ agents learn averaged responses that perform moderately against all opponent types but excel against none.”*

same, Core Problem

Definition: Two retrieval rules. Individual memory retrieves only instances recorded against the partner currently faced. Pooled memory retrieves from every partner, treating all past interactions as equally relevant.

Distinguishes it from: Category-conditional retrieval, which is the middle term: retrieve from partners sharing a category. Pooled is that middle term with one category containing everyone.

Operational test: Does the retrieval filter carry a partner index? None is pooled, an exact index is individual, a category label is the middle.

In my project: The gap is widest fully connected, where a pooled agent is averaging over the largest number of distinct partner types, and narrowest in the ring.

Compression is what people do, and it is what costs

The page argues compression from a claim about people: an agent “does not track nine separate people”, it compresses them into a small number of types, “because carrying nine independent models is not what people do”, website, Iterated Prisoner’s Dilemma, Categorization.

The network simulations find the opposite ordering on performance: individual, fully separated memory beats pooled memory on every metric in every topology, Reports/2025_0722.md.

And the identity model removes the capacity story altogether: granted perfect individuation, “the bottleneck in dense networks is not memory capacity but encounter scarcity”, Collective Intelligence manuscript, Full Information.

Three results, one direction each, and they are compatible only if compression is read as a constraint people are under rather than as a strategy that pays. That is the reading I hold, and it has a consequence the page does not state: categorization should be evaluated as a bounded-capacity account and not as a performance improvement, and any experiment that scores it on payoff alone is scoring it against a claim nobody made.

dynamic prosociality, α

“When α is high, an agent weighs the partner’s payoff alongside its own when choosing actions; when α is low, the agent acts largely in self-interest.”

Collective Intelligence manuscript, IBL Model with Identity

“ α rises when a partner exceeds expectations and falls otherwise”

same

Definition: A per-partner weight on the partner’s outcome inside the blended value, updated after every interaction from the signed gap between what was expected and what arrived.

Distinguishes it from: A social value orientation, which is a trait of the agent and one number. This is a property of a relationship, so one agent carries as many values as it has partners, and they diverge.

Operational test: Does the quantity have a partner index? A single social preference for the whole population is an orientation. One per tie is prosociality in this sense.

In my project: The corrected update is a two-branch rule on the sign of the surprise, with the positive branch floored at the current value and the negative branch capped at it, so the value is provably weakly increasing on non-negative surprise and weakly decreasing on negative surprise for any learning rate in $(0, 1]$. That property is checked by a test rather than assumed from continuity.

The printed update reverses its own stated intent

The paper states the intent in prose: the rule ensures that α increases only when a peer exceeds expectations and otherwise decreases.

Read literally against the printed gap, defined as expected minus actual, the branches are swapped: “the branch that fires when a peer *falls short* of expectation is the one with a floor that can only hold α level or raise it, and the branch that fires when a peer *exceeds* expectation has no floor and can lower α ”, ablation report, Dynamic prosociality.

Two pieces of internal evidence say this is a sign convention slipping rather than a design choice: the correction is argued in prose and never re-derived from the printed equations, and the independent reconstruction written against a related manuscript adopts the same branch pairing and then inverts its own normalisation on one branch, which only makes sense if that branch was never checked. The reconstruction implements the prose. The printed equations need the sign fixed before the paper goes out again.

1.3 Typing a partner

categorization

“It plays each partner in turn and keeps a running estimate of how often each one cooperates.”

website, Iterated Prisoner’s Dilemma, Categorization

Definition: Assigning each partner to a small set of kinds from observed behaviour, and then treating the kind rather than the partner as the unit of memory and of prosociality.

Distinguishes it from: Partner-indexed memory alone, which keeps every partner separate and shares nothing. Categorization is what lets experience with one partner reach another, and it is the only mechanism here that can act before a partner has been played.

Operational test: Does experience recorded against partner A change the decision about partner B? If not, there is memory but no categorization.

In my project: It reaches the decision in two places: category-conditional retrieval, where a partner with a stable category retrieves instances recorded against any partner sharing that category as well as its own, and category-level prosociality, where a partner with no update history yet inherits the mean value of its category on assignment.

type

“It compresses them into a small number of types”

website, Iterated Prisoner’s Dilemma, Categorization

Definition: On this page, one of three kinds a focal agent assigns to a partner from its observed cooperation rate relative to the group: cooperative, conditional, exploitative.

Distinguishes it from: The partner’s actual policy, which is not observable. A type here is a judgement made by one agent about another and can be wrong, can change, and depends on who else is in the room.

Operational test: Is the label held by an agent, or by the experimenter? An experimenter’s label is ground truth. An agent’s label is a type in this sense, and the two are scored against each other rather than confused.

In my project: Three types on the page, three named prototypes in the model, and up to nine learned categories at run time. The three names do not match, and that is the next box.

Three types, three prototypes, nine categories

The page names the types by their social reading: cooperative, conditional, exploitative, website, Iterated Prisoner’s Dilemma, Categorization.

The model matches a fingerprint by distance against named prototypes “(cooperative, defective, reactive)”, and clusters only where no confident match is found, ablation report, Behavioral fingerprinting.

And the count is not three: “The category count is capped at nine, matching the paper’s stated appeal to the upper end of Miller’s 7 ± 2 working-memory bound”, same section.

Two of the three names differ, and the page’s three is a description of the result while the model’s nine is a capacity bound. The page reads the middle type as conditional, meaning reciprocating; the model reads it as reactive, which is the same behaviour named for its mechanism rather than for its social meaning. Defective and exploitative are the same partner named from the outside and from the victim’s side. The naming has to be settled before either is taught, because the three social names are what a reader will carry away.

behavioural fingerprint

“each partner’s behavioral fingerprint is computed over a sliding window of $w = 5$ rounds using five features: cooperation rate, reciprocity, forgiveness, retaliation, and volatility.”

ablation report, Behavioral fingerprinting and category formation

Definition: The five-feature summary of one partner’s recent behaviour, over a five-round window. What the categoriser actually clusters.

Distinguishes it from: The cooperation rate on its own, which is one of the five. A partner who cooperates half the time at random and a partner who alternates share a cooperation rate and differ on volatility and reciprocity, which is what the other four features are for.

Operational test: Can two partners with the same cooperation rate land in different categories? If not, only one feature is doing work.

In my project: The five-feature specification comes from the appendix algorithms and the code, which is the specification adopted, and the narrative gloss of the same idea in the prose is treated

as an informal restatement rather than as a sixth feature set.

promotion

Definition: The event that turns a candidate cluster into a stable, named category, once membership and temporal-stability thresholds are met. A category does not exist until it is promoted.

Distinguishes it from: A prototype match, which is immediate and needs no history. Promotion is the slow route, and its thresholds are what stop a category from being invented from three rounds of noise.

Operational test: Could the label change next round on one surprising action? Then it is a match, not a promoted category.

In my project: Zero-shot transfer of a prosociality value is architecturally available only when a category arrives through an external channel before any interaction, an oracle or a reputation signal. For a category learned from the partner's own behaviour it cannot fire, because the confidence requirement guarantees the partner already has its own update history by the time the label exists. That is a property of the promotion rule, and it is reported as one rather than patched.

1.4 Contrast

contrast effect

“The categories are relative, not absolute. A partner is classified against the group mean, not against a fixed threshold, so the same behaviour lands in different categories depending on who else is in the room.”

website, Iterated Prisoner's Dilemma, Categorization

“Move a mildly cooperative partner into a generous group and they become the exploiter; move them into a harsh one and they become the cooperator.”

same

Definition: The dependence of a judgement about one partner on the distribution of the others. Identical evidence, different judgement, because the comparison set moved.

Distinguishes it from: A threshold that drifts with experience, which is a property of one relationship over time. Contrast is a property of the population at one moment, and it changes the judgement of a partner who has done nothing different.

Operational test: Hold one partner's behaviour fixed and replace the other partners. If the assigned category changes, the mechanism is contrast. If it does not, the classifier is absolute however it was described.

In my project: In the model this is not the category assignment step but the retrieval step. Activation is augmented by a stereotype score, the similarity of the observed action to the assigned category's typical behaviour, multiplied by a category-level term that grows with how far that category sits from the other active ones. Distinctive partners are attended to more.

Contrast is printed as a similarity and implemented as a distance

The printed formula for the category-level term is “a weighted cosine *similarity* between category prototype vectors”. Read literally it runs backwards: larger contrast should correspond to more dissimilar categories, not to more similar ones, and it is undefined at a zero-norm prototype, ablation report, Contrast.

The surviving code computes a weighted *distance* between the same vectors, “which is directionally consistent with the mechanism’s stated purpose”, same section.

The distance form, bounded into $[0, 1]$, is the specification adopted. The printed similarity form is kept in the codebase as a named variant, labelled as such and run only as a check, because it does not survive its own stated purpose. This is the second place in the same model where a printed equation contradicts the prose that introduces it, and both were found by reimplementing rather than by rereading.

Relative to the group, or matched against fixed prototypes

The page says the classification is relative: “A partner is classified against the group mean, not against a fixed threshold”, website, Iterated Prisoner’s Dilemma.

The implementation matches first against fixed named prototypes by distance, and falls back to clustering the currently active fingerprints only when no confident match is found, ablation report, Behavioral fingerprinting.

So the assignment step is absolute where a prototype fits and relative where none does, while the contrast term at retrieval is always relative. The page’s sentence is true of the mechanism the page is about and not of the first branch of the classifier. Either the prototype branch goes, or the page’s sentence acquires a clause.

the contrast result

“Contrast, as reconstructed to match its own stated psychological purpose, does not produce the predicted discrimination benefit in any ecology or statistical design used here, and shows a small, consistent payoff cost instead.”

ablation report, Conclusion

Definition: The finding. Contrast, correctly implemented and ablated against its own reference conditions, does not buy discrimination and costs a little payoff.

Distinguishes it from: A null from an inert implementation. The earlier draft gated the term on a condition that could never hold, so it did nothing by construction; the current one measurably changes the prosociality trajectories, and a regression test holds it to that. The null is a null of the mechanism, not of the code.

Operational test: Does enabling the mechanism change the trajectory at all? If not, the null is uninformative and the implementation is what is being measured.

In my project: Reported as it stands. The mechanism is on the page because it is the psychological claim the theory makes, and the ablation is in the report because the claim did not survive contact with its own reference conditions.

2 Working

alpha separation

Definition: The gap in prosociality value between cooperative and defective peers, held by one agent at one time. The primary diagnostic for whether discrimination is happening at all.

Distinguishes it from: Cumulative payoff, which is the behavioural outcome. They move independently, and the report treats that independence as the finding rather than reporting whichever is more favourable.

the discrimination against payoff dissociation

“category learning increases the alpha-separation mechanism diagnostic in every ecology where an alpha-separation comparison is defined, but its effect on cumulative payoff is negative or null everywhere in this collection of ecologies except the bounded-memory conditions”

ablation report, Ecology diversity

Definition: Discriminating better and doing better are separable results. So are recovering the true categories and holding a representation that helps.

Distinguishes it from: A weak effect. Both effects are present; they point in different directions, which is a different claim from either being small.

ecology

Definition: One configured population of partners, its mix of policies, its partner count, and its rounds per partner. The unit an ablation is repeated across, since a mechanism that helps in one mix can cost in another.

Distinguishes it from: A condition of the experiment. Interdependence and topology are manipulated; the ecology is the environment those manipulations sit inside, and results are reported per ecology rather than pooled across them.

bounded memory

Definition: A cap on retained instances per agent. The one family of conditions where category learning pays on cumulative outcome as well as on discrimination.

Distinguishes it from: Partner load, the number of distinct partners. Both strain the representation, and only the memory cap turns categorization into a benefit, which is what a compression account predicts.

common random numbers

Definition: Paired conditions share peer action streams and seed sequences, so a paired difference measures the mechanism net of sampling noise rather than comparing two independently noisy runs. Thirty replicate seeds throughout.

Distinguishes it from: Independent replication, which answers a different question and needs far more runs to answer it as precisely.

synthetic partner policies

“Every result in this report is generated against synthetic partner policies”

ablation report, Limitations

“not against the human multi-partner behavioral data the target paper’s own broader research program is oriented toward. The corresponding human cybersecurity-triad dataset is not available in this workspace, and no substitute dataset is presented as if it were that sample.”

same

Definition: The limit that bounds every claim in the ablation. The partners are constructed, the human triad data is not in the workspace, and no substitute is presented as if it were that sample.

Distinguishes it from: The human experiment behind the topology result, which is human data and is reported as such. The two bodies of evidence are not interchangeable and are cited separately.

3 Peripheral

tit-for-tat

Definition: Cooperate first, then copy the partner’s last action. The strategy that won Axelrod’s tournaments and the reference point every strategy in this literature is placed against.

Pointer: Axelrod’s tournaments and their nice, retaliatory, forgiving, legible summary. On the page’s animation it is the strategy whose retaliation spills across ties.

Pavlov

Definition: Win-stay lose-shift under its evolutionary name. Nowak and Sigmund showed it outperforms tit-for-tat in a population of stochastic memory-one strategies, because it corrects its own mistakes and exploits unconditional cooperators.

Pointer: Nowak and Sigmund, *Nature* 364, 56 to 58, 1993. The manuscript cites it for the mechanism behind the dyadic advantage, which is the same strategy arriving from a different literature.

interdependence theory

Definition: The decomposition of a dyadic outcome into actor control, partner control and joint control, from which the interdependence ratio is built.

Pointer: Kelley and Thibaut, and the modern restatement used in the manuscript. Worth carrying because it is the vocabulary a psychology reviewer will expect and a multi-agent reviewer will not have.

range-frequency theory

Definition: The field’s account of exactly the relative classification the page describes. A judgement is a compromise between a range principle, which spreads the categories over the

observed range, and a frequency principle, which gives each category the same number of stimuli, so a judgement of one item depends on the distribution of the others.

Pointer: Parducci. This is the citation the contrast section needs, because it says the relative classification is a documented property of human judgement rather than a modelling convenience.

the social intelligence hypothesis

Definition: The proposal that the cognitive demands of tracking many relationships, rather than the demands of the physical environment, are what selected for the size of the primate brain.

Pointer: The Machiavellian intelligence line and Dunbar's group size and neocortex result. This is where the branch index's phrase about limited computing resources comes from, and it is the argument for treating partner count as a cost rather than as an opportunity.

hidden type

Definition: In a Bayesian game, a private parameter of a player that the other players hold a belief over and update by Bayes' rule.

Pointer: The cluster's open-problems page uses the word type in this sense while this page uses it for a compressed category. The two are related and are not the same construct, and the difference is the next box.

Type, twice

On this page a type is a compression: a category one agent assigns to a partner from an observed cooperation rate relative to the group, capped in number because carrying one model per partner is not what people do.

On the cluster's open-problems page it is a latent: the cooperative and adversarial halves of the branch are described as "the same inference, recovering a hidden type from behaviour, while the other party adapts", website, Mental World Modeling, Next.

The second sentence is the one that unifies the branch, and it is worth keeping. But a compression is lossy on purpose and a latent is not, and a category assigned against the group mean is not identified in the way a Bayesian type is. If the two halves are folded into one account of how an agent types another, this is the first thing that account has to say, and it is not something either page says today.

Sociality or prosociality, on one page

The heading on the page is Dynamic Sociality. The paper title three sections below it, on the same page, is "Toward a Cognitive Theory of Interdependent Decisions in Groups: Dynamic Prosociality, Categorization, and Contrast".

One concept, two forms, in one file. Prosociality is the one carried by the model, where it names a weight on the partner's outcome; sociality would be the broader thing, and nothing in the file defines it. The paper title is the form to keep.

4 Where two locators disagree

Six entries above cite two files, or two lines of one page, that use one term differently. Each sits in the entry it belongs to rather than in a list of errata.

The disagreement	The entry it sits in
Compression as a constraint, or as something that pays	individual memory against pooled memory
The printed prosociality update against its own prose	dynamic prosociality
Three social names, three prototype names, nine categories	type
Contrast printed as a similarity, implemented as a distance	contrast effect
Classified against the group mean, or matched to fixed prototypes	contrast effect
Type as a compression, and type as a hidden latent	hidden type
Dynamic Sociality against Dynamic Prosociality	hidden type, second box

Two of these are the same problem seen twice: a printed equation that contradicts the prose beside it, found both times by reimplementing the model rather than by rereading it. That is an argument for reimplementation as a reviewing method, and it belongs in the methods note rather than here.

5 What crosses into the shared lexicon

Terms this project uses that the whole Mental World Modeling cluster uses. Their cluster sense goes in the shared file and the entries above carry only what this project fixes about them: type, opponent model, categorization, contrast, behavioural fingerprint, partner, interdependence, and typing another agent across cooperative and adversarial play.

Two of those already differ across the cluster and stay in both files. Behavioural fingerprint here is a five-feature summary of a partner’s recent play that an agent computes for itself; on the biased attacker page it is a signature an analyst could use to tell one kind of adversary from another. Opponent model here is a category assignment over that fingerprint; in the cyber world modeling cluster it is a learned embedding inferred from defender-observable signals. The difference is a finding, not a defect to be normalised away.

6 Sources

- Every quoted sentence in the Active tier is from the Iterated Prisoner’s Dilemma page and the branch index of the website, from the Collective Intelligence manuscript, from the ablation and sensitivity report, or from the IBL-Network reports. The locator on each entry says which.

- The three publications on the page are the Psychological Review manuscript under revision, the Collective Intelligence 2026 talk, and the Acta Psychologica paper on the triadic game. Definitions above come from the first two; the third supplies the triadic setting only.
- The three tiers, the three fluency levels and the four fields are from the Tech Fluency page. The reason a term whose two uses disagree belongs inside its entry is in `lexicon_method.tex`, in this folder.
- Field senses in the Peripheral tier are placed against Nowak and Sigmund 1993, Axelrod's tournaments, interdependence theory, Parducci's range-frequency theory, and the social brain literature. They are there to say what a reader from each community will already hear in the word.